

1 Archiving Concept

Overview

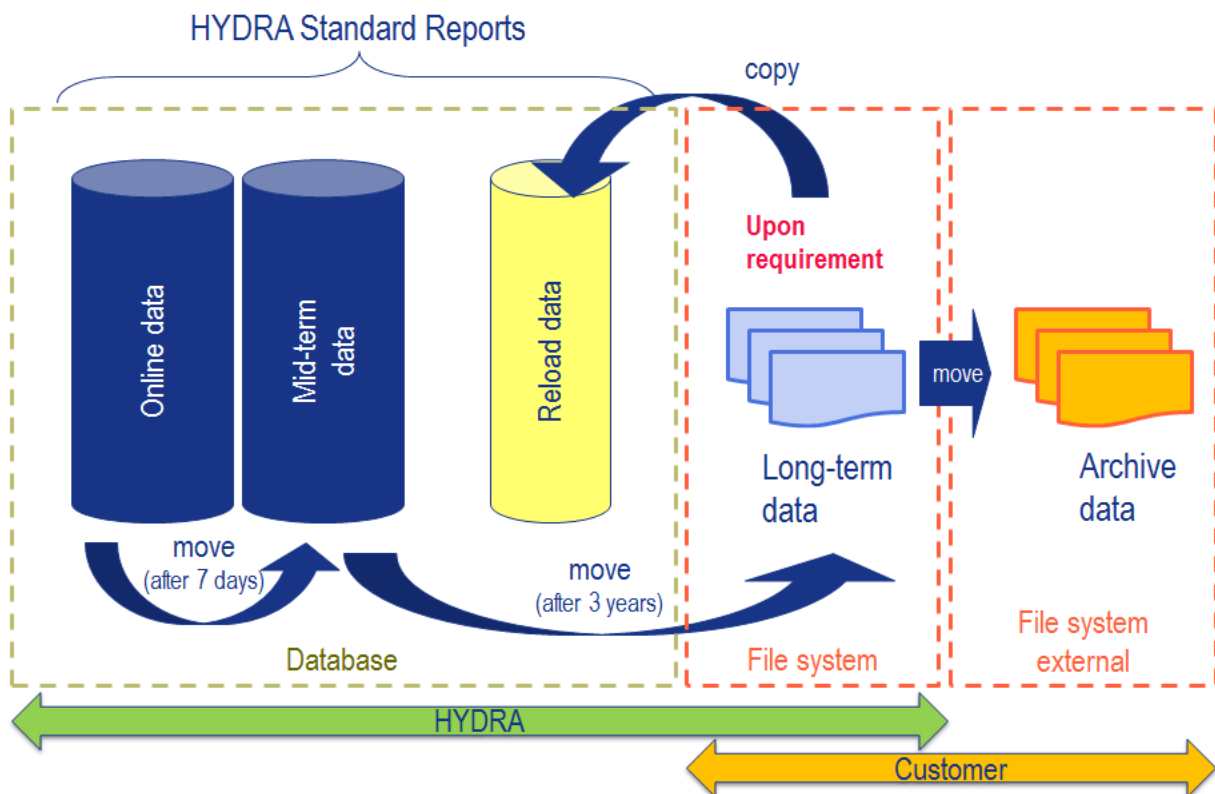


Figure 1 Data structure of archiving concept

As described in the above diagram the HYDRA data structure is divided into the following areas¹:

Online data

Online data contains all data that directly derives from HYDRA data collection and that is saved in the HYDRA database. This also includes recorded single events, log records and inventory data. This data may be edited and is available for standard evaluations.

After a configurable period of time, single events and log records are transferred to the “medium-term data” area.

¹ The retention periods indicated in the diagram are to be regarded as exemplary values as they may be defined differently in each module.

Medium-term data (archive tables)

Medium-term data contains single events, log records and stock data deriving from online data that has been moved to a separate database area after a definable period of time (may be defined separately for each module/product). Quality and granularity of data corresponds to that of the original data (no compression).

In general, this data is available for standard evaluations but can no longer be edited. The console checkbox "consider long-term data" shows whether or not long-term data is available. If this option is checked, "medium-term data" and "reload data" are included.

After a configurable period of time, this data is deleted or archived.

Long-term data

Long-term data contains single events, log records and stock data that is moved from medium-term data to long-term data after a period of time that may be defined for each module/product. (No compression).

This data is not available within the HYDRA database and may not be evaluated using HYDRA standard reports.

Data can be filed/exported in another file system (moving to archive data).

Archive data

The user is responsible for exporting long-term data to another separate area or to corresponding data carriers. HYDRA can access this data only after it has been re-imported to the long-term data area. You also require a reload to reload tables to make sure that the data is again available for long-term reports.

Reload data (reload tables)

To be able to analyze archived data from the long-term data area, the exported archive data has to be "reloaded" into the reload data of the database using the "Reload Manager". Only in this case, data is again available for the evaluations/reports that have been designed for this purpose (including long-term data function). After this reload, data is stored in a special table area, which is separated from the production data and long-term data. However, evaluations/reports are provided with this area as basis for long-term data.

This reload data may then be deleted, once the required evaluations have been performed e.g. for special researches.

Supported data classes**Events**

Events mean actually recorded single events.

Log records

This data results from event postings and respectively describes a period of time. In the booking process, the single events are usually combined to a so-called status record.

Inventory data

Inventory data has master data character but is only of temporary validity in contrast to master data.

Data archiving

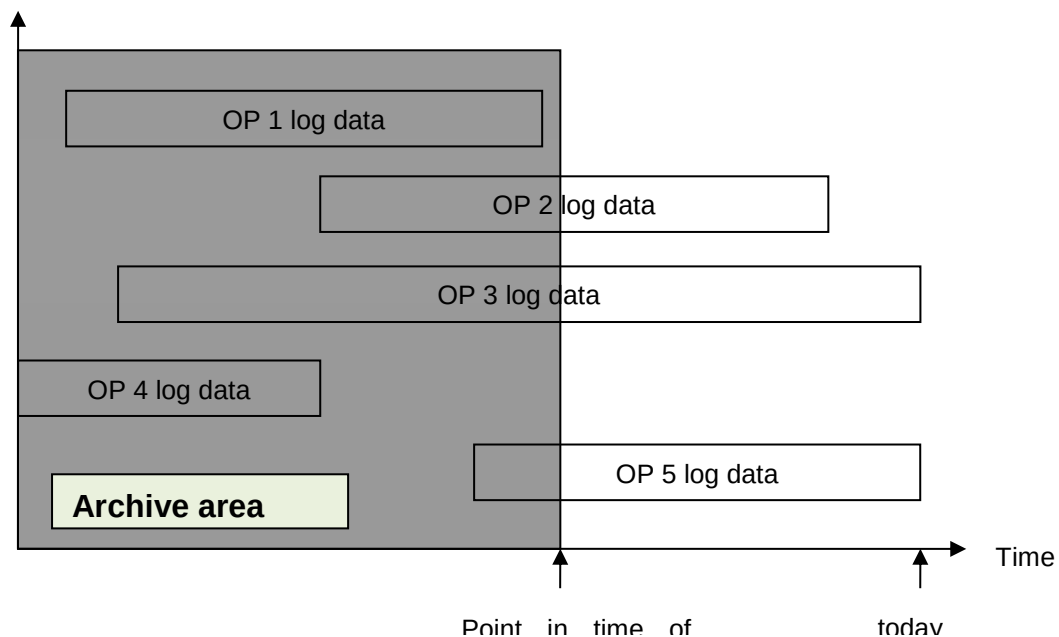
Data may be archived with respect to time or objects. To do so, archiving criteria and individual retention periods of data are configured with respect to the specific data of the single modules/products in the individual areas.

Archiving based on time

Once the defined retention period has expired, the relevant data is moved to archive tables and thus to the area of medium-term data.

After another definable retention period has expired in the medium-term data area (starts as soon as data is transferred to medium-term data), data is moved to the long-term data area. The data records of the single tables are saved in individual files that include all data records of a table for a defined period of time.

Example for archiving based on time (log data)



Archiving based on objects

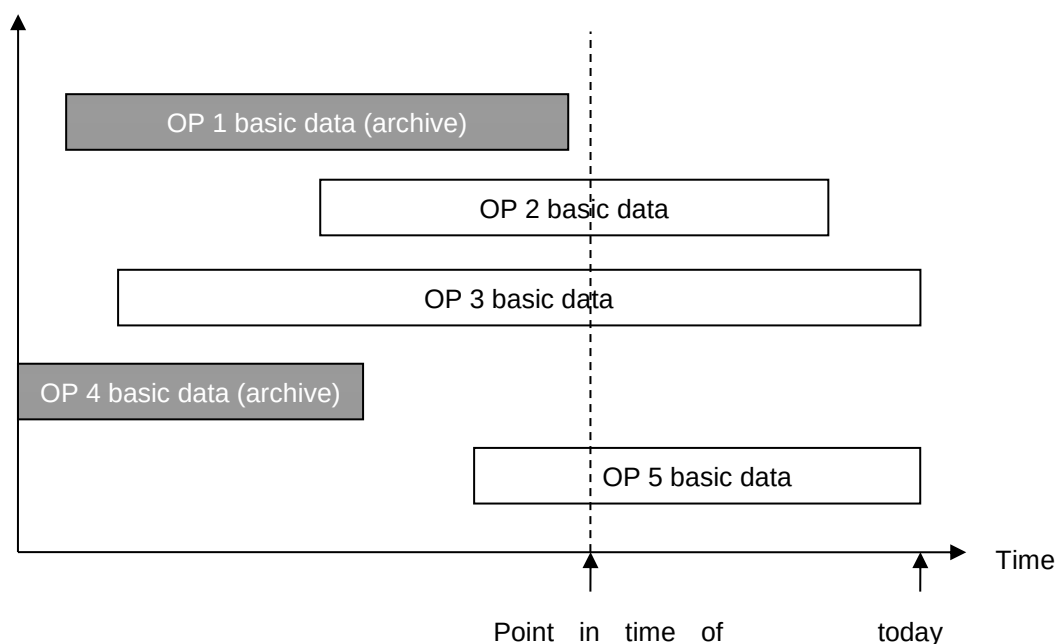
An archiving criterion that is to be defined (relating to the object, e.g. “inspection request completed”) has to be fulfilled and the retention period has to be expired to be able to archive data. Data is only archived in archive tables of the medium-term data area, once these two conditions have been met.

When the definable retention period of the single data records of an object has expired, data records are selected from archive tables and filed in object-specific files of the single tables. These files contain data records of a table of a specific object, e.g. of an order, thus a file is generated and archived for each object of the single tables.

When online data is archived in the medium-term data area (archive tables), the single data records are complemented by a field indicating the archiving time (time stamp) and a consecutive number to provide for more transparency and control. The consecutive number is very important as at the time of archiving, the same table can be archived according to several archiving criteria. In this case the file exports, which have already been generated, would be overwritten by the next export.

Within the archiving function default values for the retention period of data in the single areas, which may however be adjusted according to the customer's requirements, are defined for the individual modules (e.g. HYDRA-MDE, HYDRA-WRM, HYDRA-CAQ). For further details on this, in particular on default settings, please refer to the sections dealing with the different modules. Retention periods always refer to the date stamp of the data to be archived, i.e. with a retention period for the medium-term archive: 1 year and the long-term archive 3 years – data is archived after 1 year and two years later it is transferred to the long-term archive (those 3 years refer to the original time stamp: 1 year + 2 years = 3 years).

Example for archiving based on objects (orders)



Note when using document management

When an object is archived, also the relevant document assignments are moved to the archive. The files themselves remain at their original storage location and are not archived.